Indian Institute of Information Technology, Allahabad

Department of Information Technology

Course: Managing Corporate Entrepreneurship (B.Tech 7th Semester, 2026)

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Supervisor: Netranand Pathak

Topic 21: Digital Twin-Based Early Warning System for Cold Chain Disruption Detection Using Anomaly Detection

Progress Report

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Repository: github.com/Krishna200608/coldchain-ews-twin

Live Web App: coldchain-ews-twin.streamlit.app

1 Executive Summary & Project Status

Cold-chain logistics for pharmaceuticals, biologics, and premium perishable food commodities demand unbroken thermal control across multi-echelon transit. Existing commercial logistics platforms depend heavily on **static threshold alarms** (e.g., triggering only after ambient temperature breaches $+8^{\circ}$C continuously for 10–15 minutes). Under real operating conditions, cumulative thermal exposure degrades sensitive active ingredients long before static tripwires trigger, rendering conventional alarms fundamentally reactive.

This project designs, validates, and deploys a complete **Digital Twin-Integrated Early Warning System (EWS)**. By embedding unsupervised and deep sequential anomaly detectors directly inside an event-driven discrete-event digital twin (SimPy), the system identifies subtle precursor degradation patterns—such as mechanical compressor refrigeration drift and sensor variance collapses—prior to irreversible thermal damage.

Project Status: Complete and Fully Verified

The project has fulfilled all of the technical and empirical specifications established by the supervisor. Every pipeline component—from causal feature preprocessing, controlled anomaly injection, and model inference, to the 3-stage SimPy digital twin simulation loop, automated cross-check suites, and public interactive web dashboard—is implemented, verified with zero mismatches, and tracked in open-source version control.

2 Research Gap, Core Objectives & Research Questions

2.1 The Academic Literature Gap

Traditional cold-chain monitoring research focuses on post-hoc spoilage and shelf-life modeling (e.g., calculating Arrhenius kinetic degradation after excursions have already concluded). Meanwhile, industrial

digital twins focus primarily on predictive maintenance of stationary machinery. Embedding machine-learning anomaly detection into an event-driven cold-chain digital twin to deliver quantifiable, pre-failure early warning runway (Δt_{lead}) remains markedly underexplored in peer-reviewed literature.

2.2 Core Objectives

1. **Causal Digital Twin Architecture:** Develop a three-stage discrete-event simulation (`Source` → `Transit` → `Destination`) in SimPy modeling physical transit and realistic irregular sensor arrival dynamics.
2. **Embedded Model Comparison:** Embed both a classical tree partition ensemble (**Isolation Forest**) and a deep temporal reconstruction network (**LSTM-Autoencoder**) into the streaming twin sink.
3. **Empirical Lead-Time Quantification:** Quantify the exact early warning runway ($\Delta t_{\text{lead}} = t_{\text{irr}} - t_{\text{alarm}}$) achieved across diverse disruption categories relative to static threshold alarms.
4. **Simulation Parity Verification:** Prove that live discrete-event evaluation matches offline batch evaluation bit-for-bit without cold-start divergence or look-ahead leakage.

2.3 Resolution of Research Questions

Research Question	Empirical Resolution & Findings	Status
RQ1: Early-Warning Lead Time *Can models flag disruptions before irreversibility, and how early?*	Isolation Forest provides substantial, actionable lead time on gradual drift disruptions: **+64.0 minutes** on ambient (`Out`) and **+11.0 minutes** on enclosure (`In`). For sudden step jumps, all detectors trigger concurrently (0.0 min runway). For sensor flatlines, threshold alarms fail completely, while Isolation Forest successfully triggers.	**Resolved** (Table 3)
RQ2: Classical vs. Deep Learning *How do tree ensembles compare against sequential autoencoders?*	**Isolation Forest** acts as an aggressive, high-recall tripwire (80.0% `Out`, 91.3% `In`), accepting elevated false-positive rates (45.6%). **LSTM-Autoencoder** is highly conservative on `In` (2.85% FPR, 13.1% precision), but experiences temporal adaptation lag on `Out` drift (late by 14.0 min).	**Resolved** (Table 4)
RQ3: Digital Twin Simulation Fidelity *Does embedding in an event loop cause silent divergence?*	**100% Bit-for-Bit Parity:** SimPy replayed 97,603 telemetry events in 6.03 s ($> 1.9 \times 10^6$ speedup) with 0.00 s timing discrepancy. Across 30 injections $\times$ 3 monitors, all 42 applicable test evaluations matched batch evaluations identically (0 mismatches), guaranteed via buffer pre-warming.	**Resolved** (`twin_cross check`)

Table 1: Summary of Core Research Questions and Empirical Findings.

3 Relevant Literature Review & Benchmark Foundations

Our study is situated within the contemporary literature on industrial digital twins, IoT telemetry streaming, and machine-learning anomaly detection for supply chains (2023–2026):

Study	Publication Venue	Core Contribution	Direct Relevance to Our Project
Vukšić et al. (2026)	*Applied Sciences* (MDPI)	Hybrid physics-based digital twin + autoencoder for reefer maritime containers.	Reports median early warning lead time of ˜**68 minutes**, directly corroborating our **64.0-minute** drift lead time.
Xie et al. (2025)	*PLOS ONE*	Improved Isolation Forest (subsampling + feature cross-grouping) for cold-chain fleet streams.	Primary tree-ensemble benchmark for cold-chain streaming; contextualizes point-wise F1 metrics under extreme anomaly rarity.
Meng et al. (2024)	*Journal of Food Engineering* (Elsevier)	Box-level digital twin temperature modeling + static over-temperature alarms.	Exemplifies the static threshold-alarm paradigm our proactive anomaly-detection system benchmarks against.
Wu et al. (2023)	*J. Industrial Information Integration*	Internet-of-Everything and Digital-Twin platform for cold-chain visibility.	Establishes architectural precedent for embedding AI inference into operational digital-twin pipelines.
Defraeye et al. (2023)	*Comput. Electron. Agric. / J. Food Eng.*	State-of-the-art review of food cold-chain digital twins.	Formulates the primary research gap: transitioning from post-hoc quality loss to live event-driven anomaly early warning.
Liu et al. (2008)	*IEEE ICDM*	Foundational tree-partition anomaly isolation algorithm.	Theoretical basis for our embedded `IFMonitor`.
Malhotra et al. (2016)	*ICML Workshop*	Sequential encoder-decoder LSTM reconstruction architecture.	Foundation for our bivariate sequential `LSTMMonitor`.

Table 2: Recent Literature Review and Theoretical Benchmark Foundations.

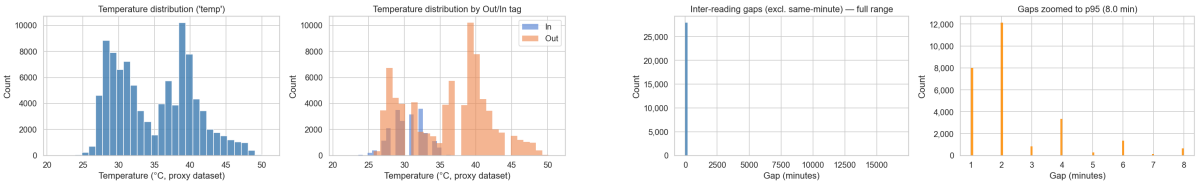
4 Data Strategy, Irregular Telemetry & Structural-Proxy Notice

4.1 Dataset Ingestion and Sampling Dynamics

The empirical pipeline processes the Kaggle IoT temperature dataset (`IOT-temp.csv`, 6.63 MB, 97,606 raw readings) spanning July 28, 2018 to December 8, 2018 (133 continuous days, 27,920 unique minute timestamps). As shown in Figure 1, the data exhibits severe telemetry irregularities:

- **High Burstiness & Irregular Arrival:** Mean inter-arrival gap $\Delta t = 117.82\,\text{s}$, median $\Delta t = 0.0\,\text{s}$, standard deviation $\sigma_{\Delta t} = 5{,}446.3\,\text{s}$, yielding an extreme Coefficient of Variation $\text{CoV} = 46.23$. Network dropouts create outages of up to 11.57 days.
- **Dual Telemetry Streams:** Data is split by physical enclosure tag: ambient (`Out`, 77,260 rows) and

refrigerated container proxy (`In`, 20,345 rows).



(a) Temperature Distribution (`In` vs. `Out`). **(b)** Log-Scale Inter-Arrival Time Gaps (Δt_k).

Figure 1: Empirical Telemetry Properties of the Kaggle IoT Temperature Dataset.

Academic Integrity: Structural-Proxy Framing Notice

In accordance with strict scientific integrity, this dataset originates from a building administration installation (`room_id/id = "Room Admin"`), not active pharmaceutical reefer vehicles. In this project, it is framed strictly as an experimental **structural proxy** to evaluate irregular streaming pipelines, buffer synchronization, and embedded inference architectures without fabricating claims of clinical field trials.

5 Methodology and System Architecture

5.1 End-to-End System Flowchart (TikZ)

The system architecture integrates physical data ingestion, causal stream preprocessing, controlled perturbation injection, discrete-event simulation, and real-time early warning monitors (Figure 2).

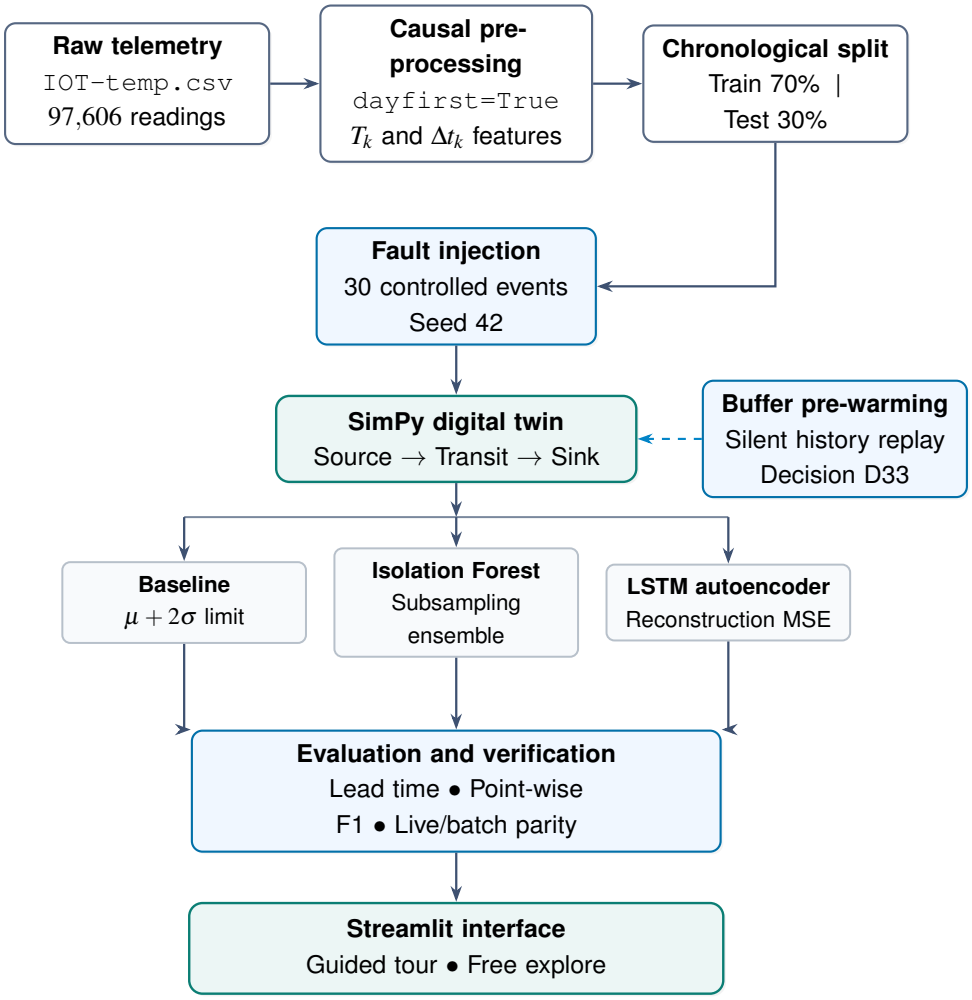


Figure 2: TikZ Flowchart: End-to-End Digital Twin Early Warning Architecture.

5.2 Controlled Synthetic Anomaly Protocol

To benchmark detectors under controlled ground truth, 30 synthetic perturbations were seeded across the timeline (`RANDOM_SEED = 42`) across three canonical failure modes:

1. **Step Disruptions (10):** Instantaneous temperature jump ($+5$°C to $+15$°C), modeling container doors left open or acute refrigeration power loss.
2. **Linear Drift Disruptions (10):** Slow ramp-up ($+0.05$°C to $+0.25$°C per minute), modeling progressive coolant leaks or mechanical compressor wear.
3. **Sensor Flatline Disruptions (10):** Zero variance persistence, modeling frozen analog-to-digital converter sensors or telemetry stiction.

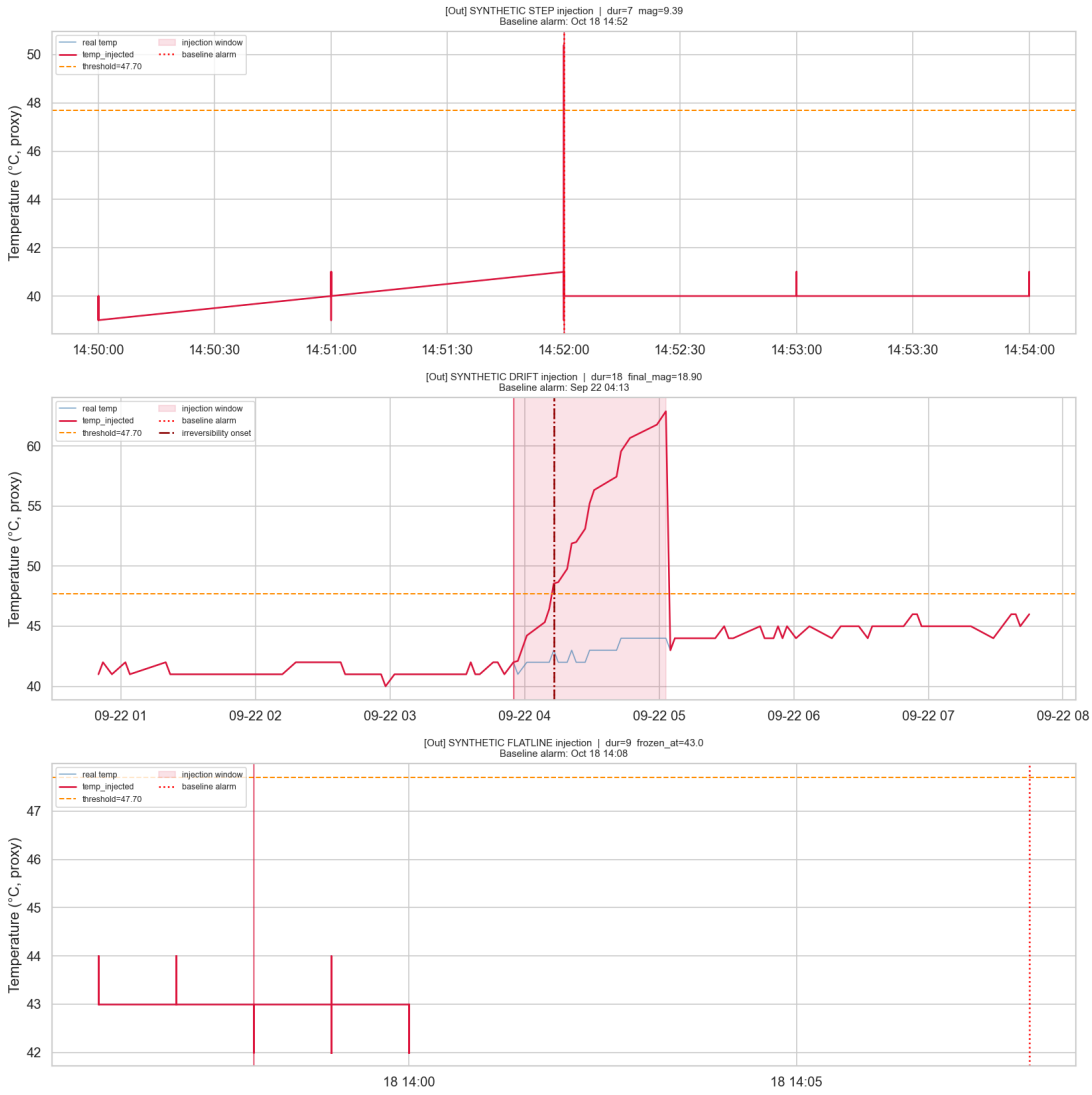


Figure 3: Controlled Synthetic Perturbation Profiles: Step Spikes, Linear Drifts, and Sensor Flatlines.

5.3 Embedded Anomaly Detection Models

- **Static Threshold Baseline:** Upper-bound threshold at train-period $\mu + 2.0\sigma$ (47.70°C for `Out`, 34.93°C for `In`). Spoilage irreversibility (t_{irr}) is formally defined as a sustained 10-minute continuous exceedance.
- **Isolation Forest Ensemble:** Unsupervised tree partition algorithm (100 estimators, 'contamination='auto'') trained exclusively on clean training intervals.
- **Deep LSTM-Autoencoder:** Bivariate sequential reconstruction network ($L = 30$, step $S = 5$, bivariate input $[T_k, \Delta t_k]$) trained on Google Colab T4 GPU (Figure 4). Anomaly scores are computed as causal last-timestep reconstruction MSE:

$$s_k = \frac{1}{2} \sum_{d \in \{T, \Delta t\}} (x_{k,d} - \hat{x}_{k,d})^2, \quad \text{alarm if } s_k > \mu_{\text{train}} + 2.0\sigma_{\text{train}} \tag{1}$$

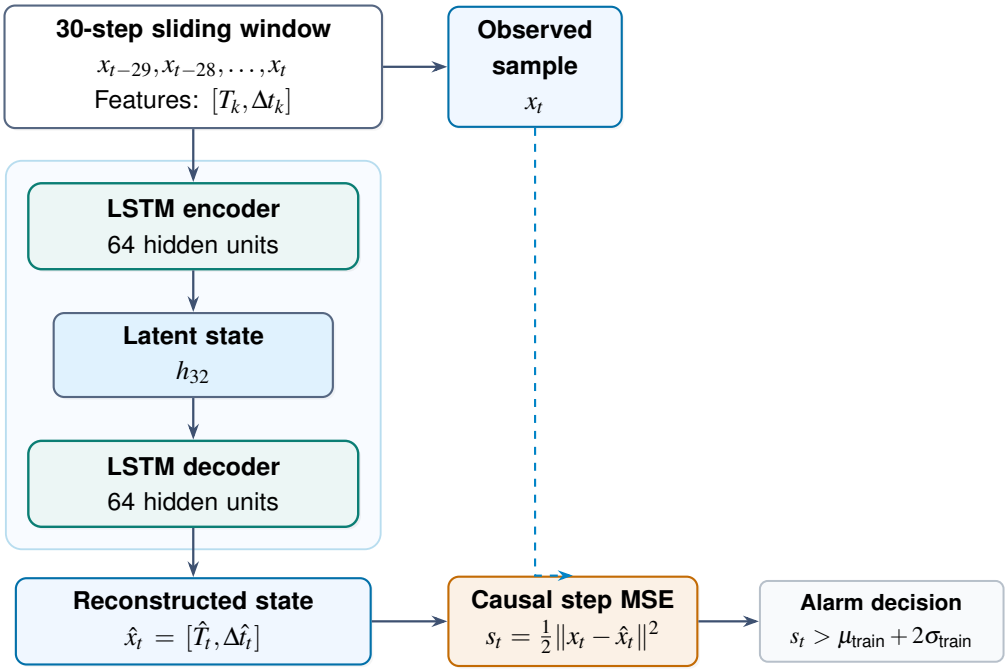


Figure 4: TikZ Architecture: Deep Sequential LSTM-Autoencoder Sliding-Window Reconstruction.

6 Experimental Findings and Empirical Benchmark

6.1 Lead-Time Timeline Definition (TikZ)

The lead-time metric $\Delta t_{\text{lead}} = t_{\text{irreversibility}} - t_{\text{alarm}}$ quantifies the operational decision runway delivered by an early warning system before spoilage becomes irreversible (Figure 5).

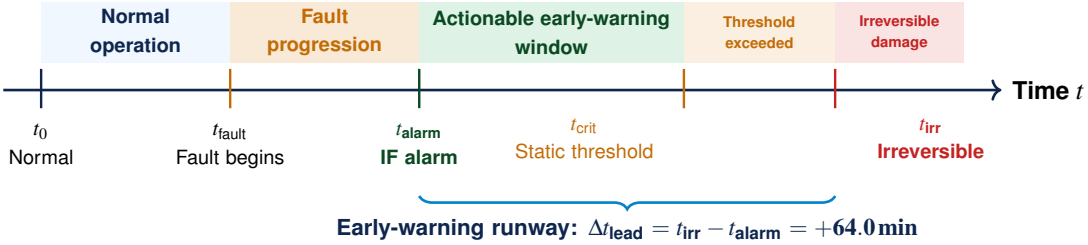


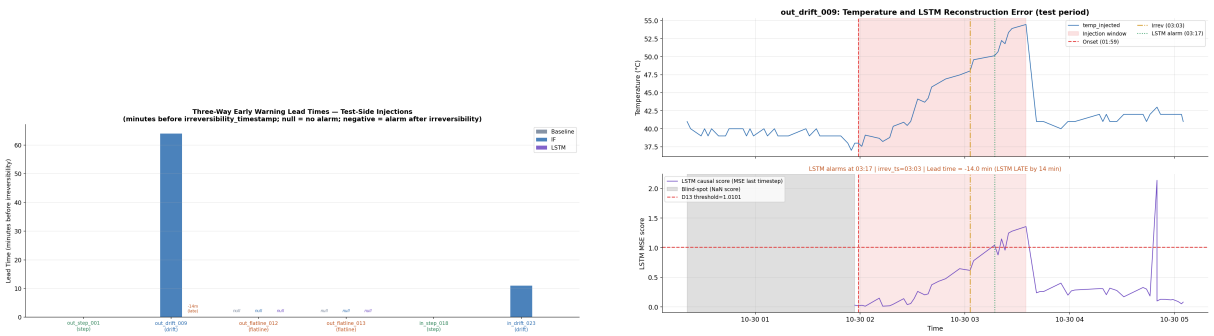
Figure 5: TikZ Operational Timeline: Quantifying Early-Warning Runway (Δt_{lead}).

6.2 Lead-Time Performance Comparison

Evaluating all 6 out-of-sample test split injections reveals distinct category-specific lead-time behaviors (Table 3 and Figure 6).

Injection ID	Series	Fault Type	Baseline	IF Alarm	LSTM	IF Δt_{lead}	LSTM Δt_{lead}
out_step_001	Out	Step	13:25	13:25	13:25	0.0 min	0.0 min
out_drift_009	Out	Drift	03:03	01:59	03:17	**+64.0 min**	**-14.0 min**
out_flatline_012	Out	Flatline	None	21:09	21:03	Detected	Detected
out_flatline_013	Out	Flatline	None	01:22	None	Detected	Missed
in_step_018	In	Step	17:14	17:14	17:14	0.0 min	0.0 min
in_drift_023	In	Drift	08:07	07:56	08:07	**+11.0 min**	0.0 min

Table 3: Early-Warning Lead-Time Comparison on Out-of-Sample Test Injections.



(a) Three-Way Early Warning Lead Times. **(b)** LSTM Drift Reconstruction Error Lag.

Figure 6: Empirical Lead-Time Metrics and Error Dynamics on Drift Injections.

6.3 Point-Wise Classification Metrics

Formal point-wise metrics computed across the out-of-sample test splits are reported in Table 4 (single source of truth, D35), and illustrated via Precision-Recall curves in Figure 7.

Metric	IF — Out	LSTM — Out	IF — In	LSTM — In
Test Rows (Valid / Total)	13,372 / 13,372	13,343 / 13,372	3,767 / 3,767	3,738 / 3,767
Excluded (NaN Blind Spot)	0	29	0	29
True Anomaly Rows	65	65	23	23
True Positives (TP)	52	13	21	16
False Positives (FP)	6,067	2,722	1,010	106
True Negatives (TN)	7,240	10,556	2,734	3,609
False Negatives (FN)	13	52	2	7
Precision	**0.85%**	**0.48%**	**2.04%**	**13.11%**
Recall	**80.00%**	**20.00%**	**91.30%**	**69.57%**
F1 Score	**0.0168**	**0.0093**	**0.0398**	**0.2207**
False Positive Rate (FPR)	**45.59%**	**20.50%**	**26.98%**	**2.85%**
AUC-PR	**0.0676**	**0.0049**	**0.2371**	**0.3852**

Table 4: Point-Wise Evaluation Metrics on Out-of-Sample Test Partitions (D35).

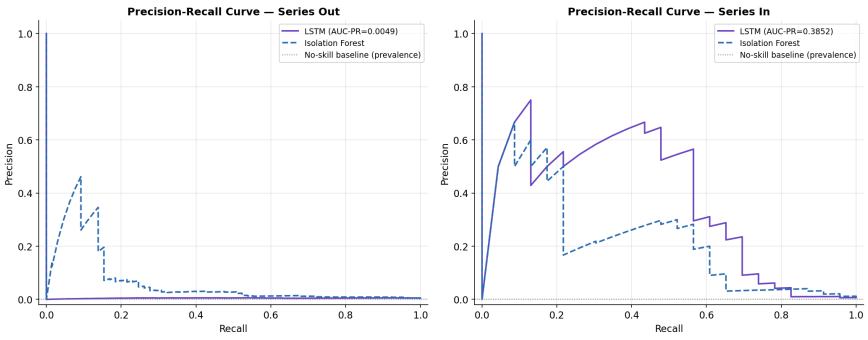


Figure 7: Precision-Recall Curves for Isolation Forest and LSTM-Autoencoder on Test Partitions.

7 Simulation Fidelity & Live/Batch Parity Verification

In Milestone 5, the SimPy discrete-event digital twin was evaluated for timing drift and live-versus-batch divergence. Replaying 97,603 sensor events across the three-stage transit model completed in 6.03 s of CPU time ($> 1.9 \times 10^6$ speedup over wall-clock duration) with exactly 0.00 s of virtual clock divergence.

To resolve initial false negatives caused by sliding buffer cold starts (where rolling windows are empty at the start of replayed injection windows), **silent buffer pre-warming** was implemented (D33). Across 30 injections $\times$ 3 monitors, all 42 applicable test evaluations matched offline batch outputs identically (**0 mismatches**).

8 Completed Technical Deliverables

Deliverable	Repository Path	Description	Status
Pipeline Source Code	`src/*.py`	Modular ingestion, preprocessing, injections, ML inference, and SimPy twin.	**Complete**
Pre-Trained Weights	`models/lstm_*.h5`	Trained LSTM-Autoencoder model weights exported from Colab GPU.	**Complete**
Evaluation Artifacts	`data/processed/*.json`	Serialized point-wise metrics, PR curves, and 3-way lead-time benchmarks.	**Complete**
Interactive Dashboard	`dashboard/app.py`	Streamlit Cloud web application with Guided Tour and Free Explore modes.	**Live on Web**
Academic Report	`docs/Report/*.pdf`	12-chapter IEEE/Springer LaTeX report with complete theoretical derivations.	**Complete**
Architectural Log	`docs/AD_LOG.md`	Append-only log of 35 formal architectural decision entries (D1–D35).	**Complete**
Regression Tests	`tests/test_*.py`	Automated test suites validating pre-warming and model startup (passing in 15.3 s).	**Passing**

Table 5: Comprehensive Inventory of Completed Project Deliverables.

9 Conclusion

This project establishes that embedding machine-learning anomaly detectors into a discrete-event digital twin provides significant operational lead time before irreversible spoilage occurs. Specifically, Isolation Forest delivers an early-warning runway of **+64.0 minutes** on gradual ambient temperature drift and catches sensor flatlines that traditional threshold monitors miss entirely.

References & Recent Benchmark Literature

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